

SIDDHARTH DEBNATH

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PROFESSIONAL SUMMARY

Data Engineer with around 7 years of experience designing and operating enterprise-scale data platforms. Engineered Kafka-based streaming pipelines processing 1.5 TB/day that improved SLA reliability and built Snowflake data models that reduced dashboard latency. Extensive experience designing AWS cloud-native architecture (EC2, ECS, Lambda, Glue, S3, Redshift, EventBridge, SNS, SQS), distributed processing, and end-to-end CI/CD automation. Proven ability to lead cross-functional data initiatives spanning pipeline engineering, data modeling, fraud analytics, and microservices development in Agile environments.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Bash (Linux Shell)

Data Engineering: Apache Kafka, Apache Spark, PySpark, Apache Airflow, Databricks, Snowflake, Redshift, Oracle, PL/SQL, PostgreSQL, Erwin Data Modeler

AWS: S3, EC2, Redshift, ECS, EMR, Lambda, Glue, CloudFormation, EventBridge, SNS, SQS, CloudWatch, IAM, VPC

DevOps / Tools: Docker, Jenkins, Git / GitHub, Artifactory, Jira, Confluence, Terraform, GitHub Copilot, Gemini

Libraries: Pandas, NumPy, SQLAlchemy, Flask, PySpark, Scikit-learn

Visualization: Tableau, New Relic

PROFESSIONAL EXPERIENCE

Capital One - Senior Data Engineer

Oct 2021 – Dec 2025

Streaming & Pipeline Engineering

- Designed and maintained 10+ Kafka-based streaming pipelines using Java and Apache Kafka to ingest data from source databases into Amazon S3 (staging) and Snowflake (target), processing 1.5 TB/day of business and web events while improving SLA reliability.
- Architected AWS-native ETL pipelines using S3 (data lake), Lambda (serverless processing), and ECS (containerized batch workloads), reducing operational overhead and improving platform scalability.
- Orchestrated ETL pipelines using Databricks Workflows with task dependencies, retries, and failure notifications for aggregating large datasets, enriching and storing the data in Delta Lake.
- Optimized job clusters in Databricks, enabling efficient data transformations and reducing resource utilization and DBU usage.
- Engineered Python and SQL queries in Snowflake for data manipulation and reporting for business users to streamline data analytics.
- Orchestrated ETL workflows using Airflow DAGs with BashOperator, PythonOperator, and Linux shell scripts for pre/post processing and operational automation.
- Created DDLs for tables, views, and materialized views in Snowflake, flattening JSON data from raw data tables for faster analytics and reporting.
- Built a Medallion Architecture analytics layer, transforming raw Bronze data into curated Silver tables and trusted Gold marts (fact and dimension models) consumed by Tableau and downstream teams.

Data Quality & Governance

- Implemented a comprehensive data quality and validation framework, combining PySpark transformations with SQL-based checks (null checks, schema drift detection, and referential integrity) across all curated datasets.

- Designed schema validation mechanisms to enforce backward/forward compatibility and prevent schema drift across downstream consumers.
- Worked on role-based access control for highly sensitive data using different views on top of base tables.
- Enhanced data encryption for PII/PCI information using AES-256 and KMS for encryption/decryption.
- Leveraged AI tools like GitHub Copilot and Gemini for code development, enrichment, understanding existing codebases, and catching errors.
- Established data lineage and metadata documentation using Erwin Data Modeler, enabling faster data discovery and reducing the time to insight for analytics teams.
- Partnered with product, analytics, and risk teams to align on data quality, compliance, and lineage standards, resulting in measurably fewer data-related audit findings.
- Developed framework for detecting highly sensitive data from agent notes using regex and masking/removing them for RBAC.

Fraud Analytics & Financial Domain

- Reviewed the architectures and code for loan application pre-screening systems, ingesting and normalizing applicant identity and credit bureau data from TransUnion, Experian, and Equifax.
- Served as the first line of defense in fraud detection pipelines, identifying high-risk patterns including duplicate SSNs, identity mismatches, and conflicting applicant attributes across systems.
- Conducted deep-dive analysis on financial datasets from Snowflake, Oracle, Salesforce, and Redshift using Pandas and Spark, generating insights that guided enterprise risk assessments and audit reviews.

Microservices & Application Development

- Developed and deployed microservice-based REST APIs using Python (Flask) integrated with Snowflake, Pandas, and AWS services, exposing curated datasets for internal applications and analytics consumers.
- Collaborated with frontend teams to define API contracts and data schemas, reducing integration friction and accelerating feature delivery cycles.
- Authored technical documentation including API specifications, data dictionaries, workflow diagrams, and runbooks in Confluence, improving onboarding time and operational support quality.

CI/CD & Platform Operations

- Designed and maintained CI/CD pipelines using Jenkins, Artifactory, GitHub and Linux Bash scripts to automate builds, deployments, artifact packaging, and environment configuration.
- Containerized ETL jobs and backend services using Docker, deploying to AWS ECS and Lambda for scalable, repeatable execution across environments.
- Implemented Infrastructure as Code using Terraform to provision data infrastructure, networking, IAM roles, and monitoring resources.
- Configured CloudWatch alerts, logs, and dashboards for end-to-end pipeline observability; managed production incidents via ServiceNow, improving mean time to acknowledgment and reducing repeat failures.

Capital One — Data Engineer

Aug 2019 – Sept 2021

Pipeline Development & Integration

- Developed Python ETL frameworks using PySpark and AWS Glue to ingest, validate, and transform large-scale datasets from Oracle, Snowflake, and REST APIs.
- Engineered data models (star and snowflake schemas) in Snowflake using Erwin Data Modeler with performance-tuned clustering, reducing dashboard query latency from 16s to 3.5s.
- Engineered high-performance data pipelines on PySpark, optimizing data ingestion, transformation, and aggregation to ensure reliable downstream analytics and reporting.
- Migrated legacy Oracle databases using Pandas + SQL ETL jobs orchestrated with Airflow DAGs and Docker, achieving 99.9% data integrity across all migrated datasets.

- Wrote and optimized complex SQL queries on large, distributed datasets, leveraging partitioning, clustering, and query plan analysis to reduce latency and improve resource efficiency.

Analytics & Reporting

- Developed interactive Tableau dashboards combining data from Snowflake, Redshift, Salesforce, and Excel, enabling stakeholders to monitor MBR and daily reporting of key business metrics.
- Implemented SQL-based rollup reports and defined business KPIs for cross-functional stakeholders, improving visibility into performance trends.
- Automated Jira project tracking dashboards in Tableau, improving Agile sprint visibility across engineering and product teams.
- Developed dashboards for UI interactions for users in New Relic to monitor network traffic, response time, latency, and usage paths.

Infrastructure & Testing

- Containerized ETL jobs using Docker and deployed them to AWS ECS and Lambda for scalable batch execution.
- Built Terraform templates for infrastructure provisioning across development and production environments.
- Wrote unit and integration tests for Python data pipelines using pytest, ensuring correctness across schema changes and upstream dependency updates.
- Configured CloudWatch alerts and logs to monitor pipeline health, latency, and failures, improving platform availability and incident response time.

UT Dallas — HBSS Lab — Researcher

Aug 2018 – April 2019

- Developed deep learning models (CIFAR-10, YOLO, Faster R-CNN) and built custom-labeled datasets for object detection.
- Benchmarked detection algorithms for accuracy and performance for robotics use case.
- Integrated vision and control pipelines for autonomous humanoid robot tasks.

IIT Jodhpur — Researcher

June 2016 – March 2017

- Implemented SVD and DWT algorithms in MATLAB for a hybrid image-watermarking method.
- Evaluated multiple combinations for optimal robustness and minimal image distortion.

EDUCATION

The University of Texas at Dallas — M.S., Electrical Engineering

May 2019

Maulana Abul Kalam Azad University of Technology — B.Tech., Electronics & Communications Engineering

May 2017